

# Schematic

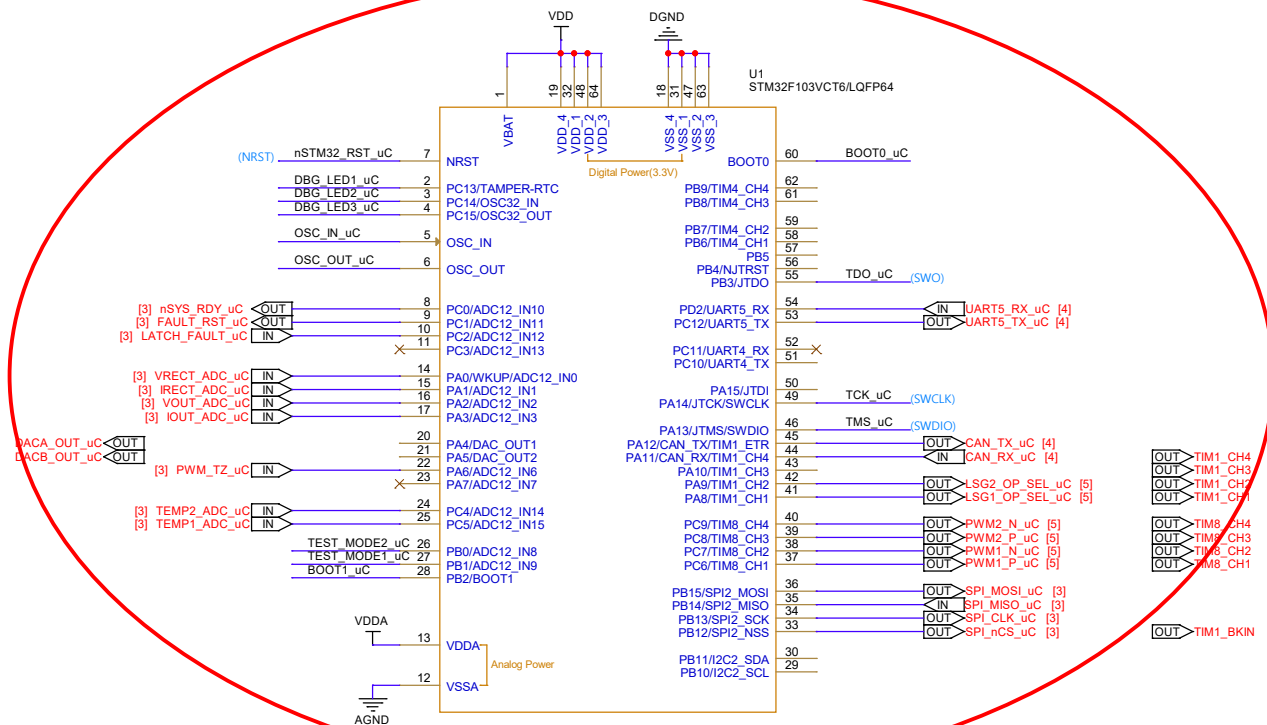
(RX OLED Regulator Control Board, Ver 0.1E00)

Date : 2026.03.27  
Design by : Hyun-min, Lee

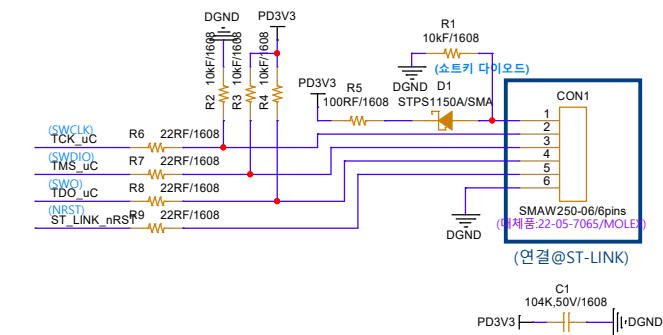
Eta Electronics Inc.

| Reference                                                                           | Description | DSN.BY     | CHK.BY     | CHK.BY     | APPR.BY    | MODEL                                    | OLED700 WPT              |               |
|-------------------------------------------------------------------------------------|-------------|------------|------------|------------|------------|------------------------------------------|--------------------------|---------------|
|  |             | H.M.LEE    | C.H.KA     | B.H.CHOI   | T.D.YEO    |                                          |                          |               |
|  |             |            |            |            |            | Title<br>Rx OLED Regulator Control Board |                          |               |
|  |             |            |            |            |            | Size<br>A3                               | Document Number<br><Doc> | Rev<br>0.1E00 |
|                                                                                     |             | 2026.03.27 | 2026.03.27 | 2026.03.27 | 2026.03.27 | Date: Friday, March 27, 2026             | Sheet 1 of 7             |               |

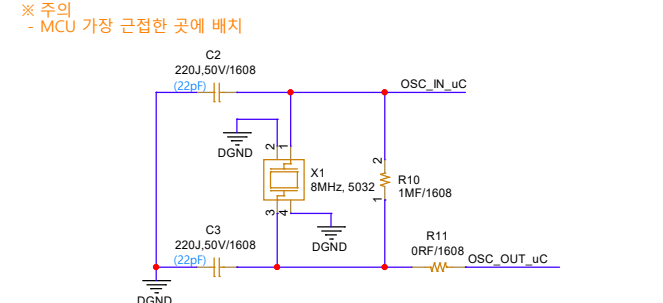
STM32 Pin Assign



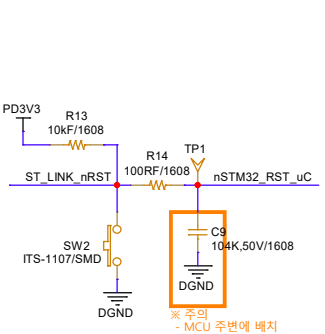
커넥터 (ST-LINK SWD)



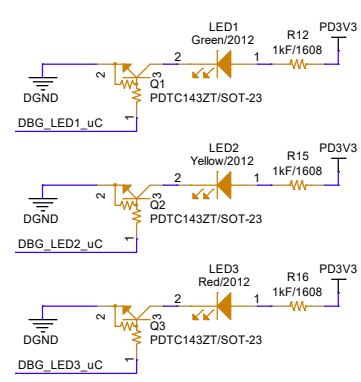
OSC Clock (8MHz)



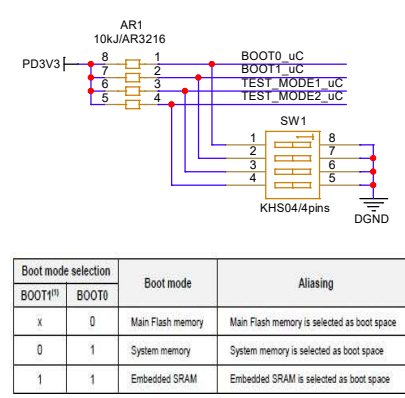
System Reset 회로부



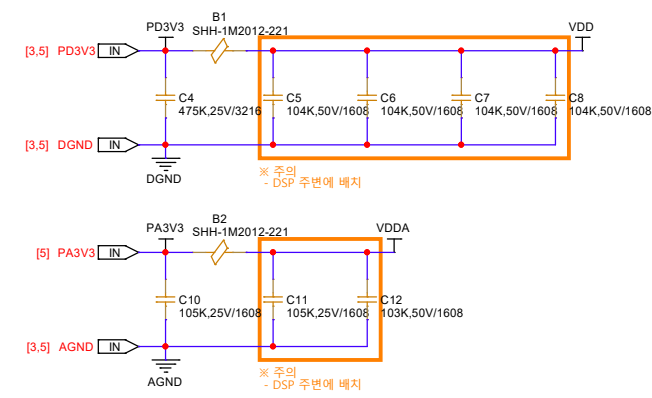
디버깅용 LED Indicator



Boot Mode Selection



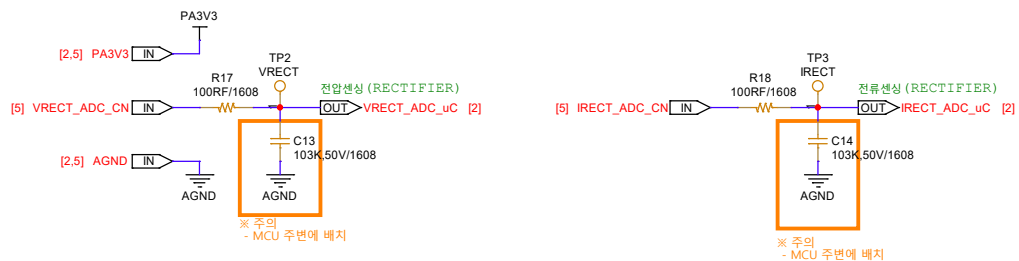
Bypass Capacitor (MCU 근처에 배치)



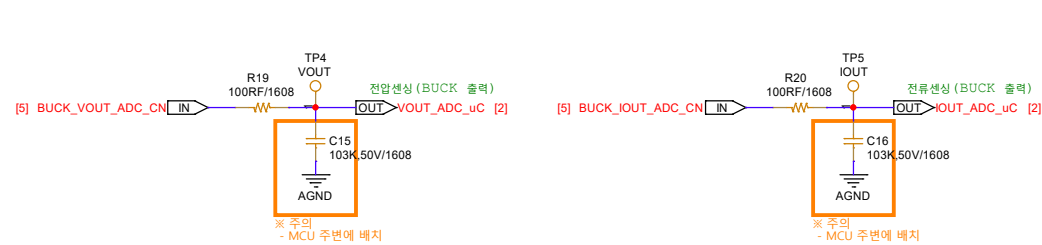
| Reference | Description | DSN.BY     | CHK.BY     | CHK.BY     | APPR.BY    | MODEL | OLED700 WPT |
|-----------|-------------|------------|------------|------------|------------|-------|-------------|
| 1         |             | H.M.LEE    | C.H.KA     | B.H.CHOI   | T.D.YEO    |       |             |
| 2         |             | 2026.03.27 | 2026.03.27 | 2026.03.27 | 2026.03.27 |       |             |
| 3         |             |            |            |            |            |       |             |

|       |                        |                                 |        |
|-------|------------------------|---------------------------------|--------|
| Title |                        | Rx OLED Regulator Control Board |        |
| Size  | Document Number        | Rev                             |        |
| A3    | <Doc>                  | D.1E00                          |        |
| Date  | Friday, March 27, 2026 | Sheet                           | 2 of 7 |

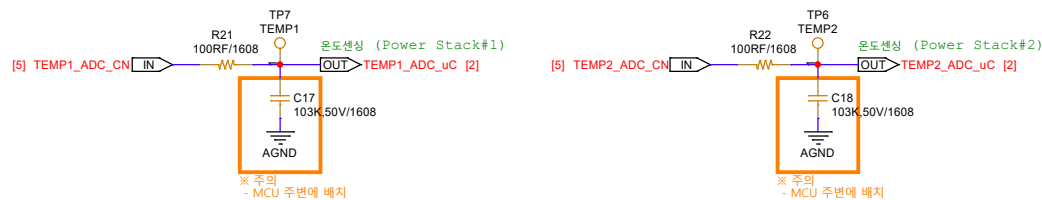
## ADC LPF (Rectifier) : Connector to uC



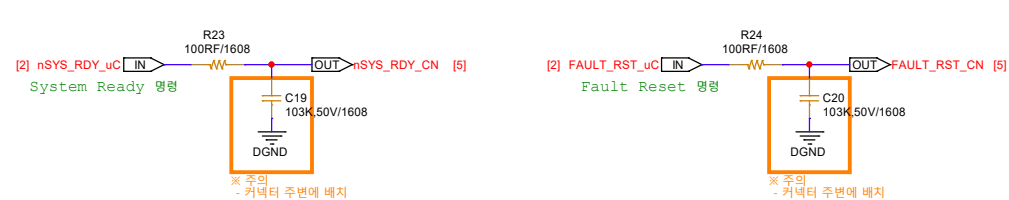
## ADC LPF (Buck Output) : Connector to uC



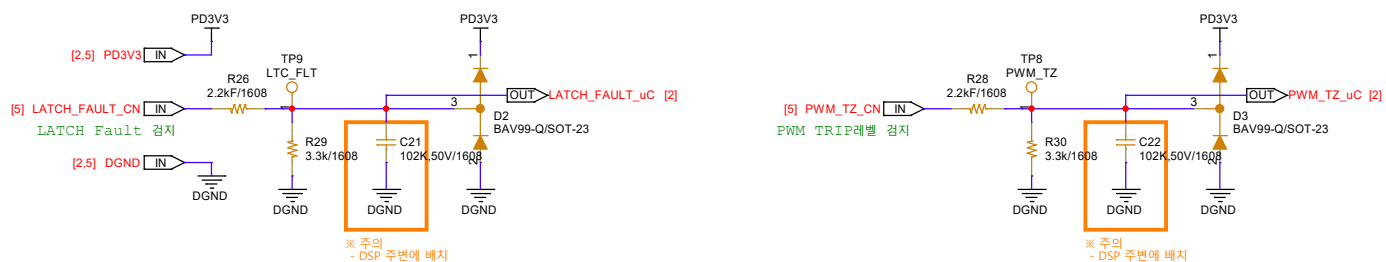
## ADC LPF (방열판 온도검출) : Connector to uC



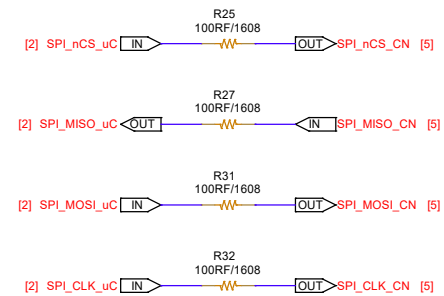
## LPF(Command for System Operating) : uC to Connector



## LPF(Detection for System Status) : Connector to uC



## SPI Comm Signal

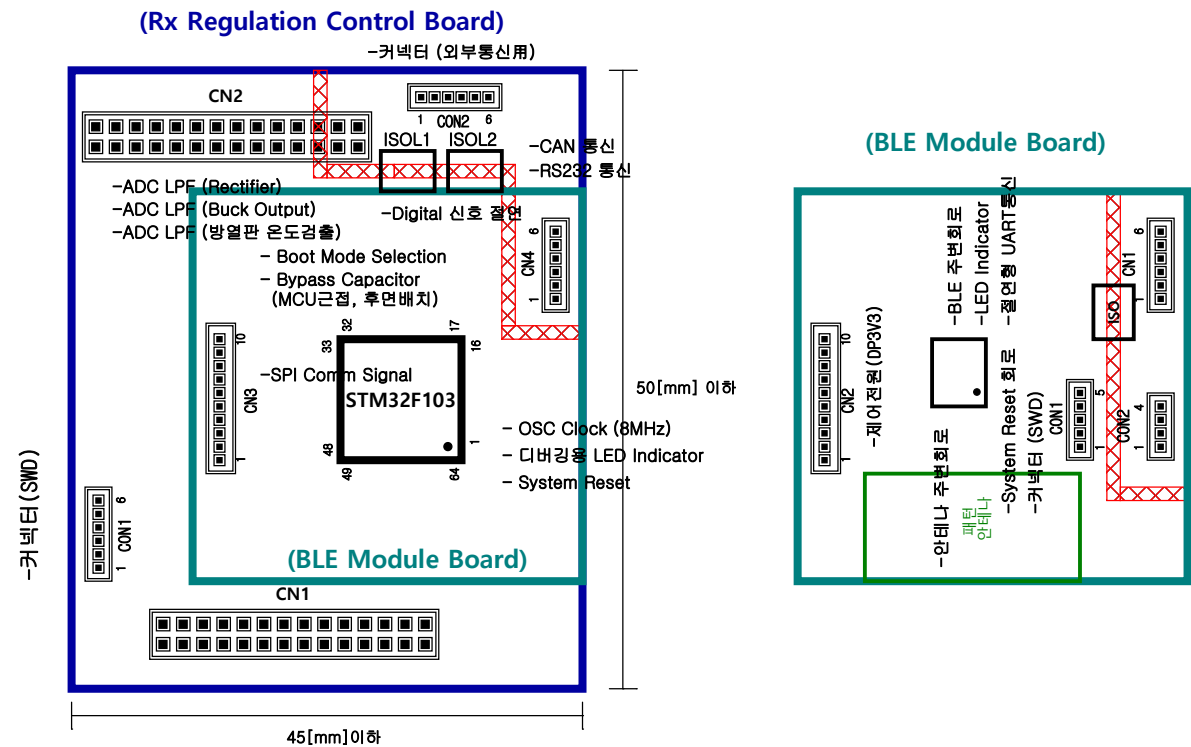


| Reference | Description | DSN.BY     | CHK.BY     | CHK.BY     | APPR.BY    | MODEL                                                                                                                  | OLED700 WPT |
|-----------|-------------|------------|------------|------------|------------|------------------------------------------------------------------------------------------------------------------------|-------------|
| 1         |             | H.M.LEE    | C.H.KA     | B.H.CHOI   | T.D.YEO    | Title Rx OLED Regulator Control Board<br>Size Document Number<br>A3 <Doc><br>Date: Friday, March 27, 2026 Sheet 3 of 7 |             |
| 2         |             |            |            |            |            |                                                                                                                        |             |
| 3         |             | 2026.03.27 | 2026.03.27 | 2026.03.27 | 2026.03.27 |                                                                                                                        |             |





Rx Regulator Control Board 부품배치도



| Reference | Description | DSN.BY     | CHK.BY     | CHK.BY     | APPR.BY    | MODEL                                 | OLED700 WPT           |
|-----------|-------------|------------|------------|------------|------------|---------------------------------------|-----------------------|
| 1         |             | H.M.LEE    | C.H.KA     | B.H.CHOI   | T.D.YEO    | Title Rx OLED Regulator Control Board |                       |
| 2         |             |            |            |            |            | Size A3                               | Document Number <Doc> |
| 3         |             |            |            |            |            | Date: Friday, March 27, 2026          | Rev 0.1E00            |
|           |             | 2026.03.27 | 2026.03.27 | 2026.03.27 | 2026.03.27 | Sheet 6 of 7                          |                       |

Rx OLED Regulator Control Board의 Revision 내용

1. Rx OLED Regulator Control Board (Ver0.1E00 / 생성일자 : 2026.03.27)
- 회로도 초기 생성

설계 참조자료

TEST PIN 처리 설명

| 모양                                                                                  | 내용설명            |
|-------------------------------------------------------------------------------------|-----------------|
|  | Via Hole 처리     |
|  | Test Pointer 처리 |

캐패시터 온도 및 오차

| CODE | 오차         | 온도특성       |
|------|------------|------------|
| COG  | ±30[ppm]   | -55~125[℃] |
| X7R  | ±15[%]     | -55~125[℃] |
| Y5V  | ±22~-82[%] | -30~85[℃]  |

칩저항 SIZE

| Type           | Power    | 최대사용전압 |
|----------------|----------|--------|
| 0803<br>(0201) | 0.05[W]  | 25[V]  |
| 1005<br>(0402) | 0.1[W]   | 50[V]  |
| 1608<br>(0603) | 0.1[W]   | 50[V]  |
| 2012<br>(0805) | 0.125[W] | 150[V] |
| 3216<br>(1206) | 0.25[W]  | 200[V] |
| 3225<br>(1210) | 0.5[W]   | 200[V] |
| 5025<br>(2010) | 0.75[W]  | 200[V] |
| 6432<br>(2512) | 1.0[W]   | 200[V] |

칩저항 오차

| Type | 오차       |
|------|----------|
| B    | ±0.1[%]  |
| C    | ±0.25[%] |
| D    | ±0.5[%]  |
| F    | ±1.0[%]  |
| G    | ±2.0[%]  |
| J    | ±5.0[%]  |
| K    | ±10.0[%] |
| M    | ±20.0[%] |

 : PCB ROUTER 0.5mm...0.8mm

| Reference                                                                           | Description | DSN.BY     | CHK.BY     | CHK.BY     | APPR.BY    | MODEL                           | OLED700 WPT            |        |
|-------------------------------------------------------------------------------------|-------------|------------|------------|------------|------------|---------------------------------|------------------------|--------|
|  |             | H.M.LEE    | C.H.KA     | B.H.CHOI   | T.D.YEO    | Title                           |                        |        |
|  |             |            |            |            |            | Rx OLED Regulator Control Board |                        |        |
|  |             |            |            |            |            | Size                            | Document Number        |        |
|                                                                                     |             | 2026.03.27 | 2026.03.27 | 2026.03.27 | 2026.03.27 | A3                              | <Doc>                  | 0.1E00 |
|                                                                                     |             |            |            |            |            | Date:                           | Friday, March 27, 2026 |        |
|                                                                                     |             |            |            |            |            | Sheet                           | 7 of 7                 |        |